

Adaptive Channel Access Mechanism for Real Time Traffic Over IEEE 802.11e Wi-Fi Network

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Abstract - The deployment of Wi-Fi network has been growing rapidly due to its ease of implementation and affordability in prize. The increase in number of users and applications results in increment of multimedia traffic in Wi-Fi networks. There are lots of performance issues in Wi-Fi network with respect to Quality of Services (QoS), especially with voice and video applications. In this paper we propose the adaptive channel access mechanism to improve the performance of real time traffic under high traffic environment. The proposed adaptive channel access mechanism improves the performance of voice and video traffic by 15% to 20% in terms of packet delay and throughput under high traffic load environment.

Keywords-EDCA; HCF; Wi-Fi; QoS; VoIP

I. INTRODUCTION

The 802.11 wireless local area networks (WLANs) has been getting popular in network deployments mainly because of their ease of implementation and usage. In addition, the equipments are also cheaper. From the user point of view, they function and perform more like a shared Ethernet LAN. The IEEE 802.11 WLANs could be installed both in infrastructure and ad-hoc mode. The infrastructure mode is widely used as an access network called Wi-Fi or Hotspots due to their ease of implementation and affordability. The IEEE 802.11 was the first release of the WLAN standard which did not have any service differentiation. Hence, it provided best effort services for all the types of traffics. With the increase of WLAN users and applications, there is an increase of real time traffic in WLANs. Since IEEE 802.11 does not have any Quality of Service (QoS) guarantee for different traffic categories, there are performance issues with multimedia applications (VoIP and Video) in WLAN [1].

The IEEE 802.11e was included in the IEEE 802.11-2007 release to support QoS for various traffic categories (AC). The 802.11e station has four different Access Categories. They are categorized as voice AC[0], video AC[1], best effort AC[2] and background traffic AC[3], where AC[0] is the highest and AC[3] is the lowest priority traffic category, respectively. Even though enhanced distributed channel access (EDCA) in IEEE 802.11e has its own service differentiation method, it could not provide any QoS guarantees for the real time multimedia applications

(VoIP and Video Conferences) under high load conditions [1]. There are lots of researches trying to improve the QoS of real time applications in IEEE 802.11e but most of the works focused on the uplink transmission (Stations to Access Point). Only few works were carried to enhance the downlink transmission (Access Point to Stations), which suffers due to heavy downlink traffic to multiple stations. In IEEE 802.11e EDCA mode, the AP (Access Point) also has to compete for the channel like the other stations in the BSS (Basic Service Set –including AP and Stations in Wi-Fi [2]). The real time multimedia applications such as videos online, VoIP, Video conferencing and real time applications based on TCP protocol have more downlink traffic with respect to the Uplink traffic.

In 802.11e EDCA the Access Point can seize the channel with the same probability as the other stations and usually needs to transmit a huge amount of data, then, a downlink bottleneck is predictable. Since the VoIP and Video applications have a huge amount of data flow in the downlink, their performances get degraded. The delay and packet loss are comparatively high in the downlink due to the bottleneck in the downlink [2]. Since the same service category in all stations has the same EDCA parameters (AC[0] for voice which has the same CW_{min} , CW_{max} and transmission opportunity (TXOP) for all the stations), there will be lots of collision, which results in performance degradation when all stations have VC[0] traffic to transmit. In the scenario when all stations have video application (VC[1] traffic category) and if some stations have more than one video flow, there will be delay in packet transmission in those stations having more flows. Since the EDCA parameters are identical for the particular traffic category irrespective of the number of flows, which leads to packet delay and performance degradation for real time applications.

In this paper, we propose the adaptive channel access mechanism for real time traffic over IEEE 802.11e to improve the performance of voice and video applications under high traffic environment. This standard is completely complied with the IEEE 802.11e draft standard. The proposed mechanism could sense the load of voice and video traffic by measuring their queue length and dynamically adjust the contention window parameters in order to improve their performance. The rest of this paper is

organized as follows: Section II reviews the IEEE 802.11e standard. In section III, we discuss the proposed adaptive channel access mechanism in detail. Simulation results are presented in Section IV. Section V concludes the paper.

II. OVERVIEW OF IEEE802.11E AND RELATED WORKS

The IEEE standard 802.11e provides QoS in two forms [4]. Firstly, it supports a priority-based service, which is similar to DiffServ. Secondly, it supports parameterized QoS for the benefit of applications requiring QoS for different flows. In IEEE802.11e, this can be achieved by enhancing the distributed coordination function (DCF) and point coordination function (PCF), and by providing a signaling mechanism for parameterized QoS.

A. Hybrid Coordinate Function (HCF)

The HCF includes two channel access mechanisms: EDCA and HCCA. The EDCA (Enhanced Distributed Channel Access) is the contention and the distributed access method. It is an enhancement of DCF and is used both in ad-hoc mode and in Wi-Fi mode. The HCCA (Hybrid Controlled Channel Access) is the method controlled by hybrid coordinator (HC). It is used only in infrastructure network because of its centralized control fashion. IEEE 802.11e includes also other new features to provide better QoS performance. A TXOP (transmission opportunity) is a limited time interval reserved for a specific station. If the frame length is smaller than the TXOP, the station is permitted to send as many frames as it can during its TXOPs. If the frame length is more than the TXOP, the station must split the large frame into smaller blocks where each of which can be transmitted in the length of TXOP. The introduction of TXOP reduces the problem of station transmitting at low rate gaining an inordinate amount of channel time in the 802.11 legacy DCF.

B. Enhanced Distributed Channel Access (EDCA)

In the IEEE 802.11e EDCA, traffic is classified into four access categories AC_i ($i = 0, 1, 2, 3$), each of which follows a carrier sense multiple access and collision avoidance (CSMA/CA) mechanism. When AC_i has a data frame to transmit, it first senses the wireless link for an arbitrary inter-frame space (AIFS[i]) period given by $AIFS[i] = SIFS + AIFSN[i] \times SlotTime$, where SIFS, AIFSN[i], and SlotTime denotes the shortest possible inter-frame space, an integer greater than one, and the duration of a slot, respectively. When $AC[i]$ finds the wireless link idle for an AIFS[i] period, the random back-off procedure starts. $AC[i]$ generates a random back-off interval as an additional deferral time before starting its frame transmission and the back-off counter decreases while the wireless link is being idle. Note here that the back-off counter is frozen when the wireless link is sensed busy, and it is reactivated when the wireless link is sensed idle again for an AIFS[i] period. When the back-off counter reaches zero, $AC[i]$ tries to access to the wireless link. If back-off counters of more than one access category in a specific station expire at the same time slot, so-called an internal collision occurs. The internal

collision is resolved within the station using virtual collision handler, as shown in Fig. 1, the frame of the access category with highest priority is transmitted and others behave as if they underwent an external frame collision on the wireless link.

The random back-off interval is a random integer multiplied by SlotTime and it is determined by each access category individually. The random integer at AC_i ($i = 0, 1, 2, 3$) follows a uniform distribution on $[0, CW[i]]$, where $CW[i]$ is referred to Contention Window of AC_i . $CW[i]$ is initially set to be $CW_{min}[i]$ and for the k^{th} ($k = 1, 2, \dots, k_{max}$) retransmission, it is given by

$$CW[i] = \min((CW_{min}[i] + 1) - 1, CW_{max}[i]) \quad (1)$$

Here, k_{max} denotes the maximum number of retransmissions.

If the retransmission of $AC[i]$ fails k_{max} times successively, the frame is dropped. When a frame succeeds in transmission or it is dropped, $CW[i]$ is reset to be $CW_{min}[i]$. The IEEE 802.11e EDCA utilizes positive acknowledgements (ACKs), and the receiver sends an ACK frame within a SIFS period after the correct receipt of the transmitted data frame. If the sender fails to receive the ACK frame, it tries to retransmit the data frame. Also, once an $AC[i]$ acquires the transmission right, it allows sending multiple data frames within the duration of TXOP[i].

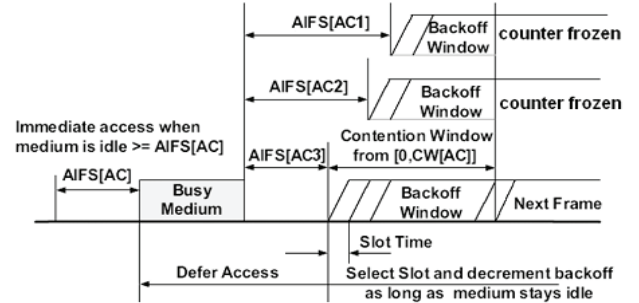


Figure 1. The medium access sequence in EDCA [3]

C. Related Work

Although the IEEE 802.11e has QoS to support various traffic categories, it is necessary to protect real time traffic under high load. There are lots of works focusing on QoS improvement for voice and video traffic. The numerical results in [2] showed that AIFS could provide substantial discrimination within the different ACs. In [3], the throughput of IEEE 802.11 DCF and IEEE 802.11e EDCA are compared. It concluded that EDCA CFB could produce better results in overall throughput. According to [4], data dropped rate is significantly reduced if Contention Free Burst (CFB) is adopted. This is because CFB could mitigate contention. The discussion of the impact of certain parameters, such as frame size and mobility, on the system throughput was carried out in [5]. Based on the simulation results, even though the mobility of one station may affect the throughput in DCF, EDCA CFB would not degrade the performance of EDCA so much. The analytical models of

EDCA were presented in [6] and [7]. They targeted on the saturation throughput analysis. The Virtual Group was introduced in [8]. Traffic is divided into various Virtual Groups. A station is only allowed to contend with those in the same Virtual Group. This paper investigates the delay and throughput performance.

III. PROPOSED CONTENTION ADAPTION

A. Adaptive Channel Access Mechanism for Real Time Traffic

Even though the IEEE 802.11e EDCA could successfully guarantee the statistical priorities of various ACs, it still confronts severe deterioration in performance while traffic load is heavy. Because EDCA is contention-based, the stations suffer from serious collisions in high traffic load. Meanwhile, energy consumption, which is critical for portable devices, is dissipated due to frequent retransmissions. Furthermore, it may be hard to meet the rigorous delay requirement of real time applications. For example, the one-way delay bound of voice traffic is 150 ms, and the loss rate must be less than 5% [9]. Based on our simulation study, EDCA could not totally assure the QoS requirements. Most of the research presented in Section II C focuses mainly on the analysis of EDCA performance. However, only few of them aim to enhance the 802.11eEDCA based on the perspective of contention alleviation. In addition, none of them considers the energy conservation along with QoS guarantee.

Our proposed mechanism could be incorporated into IEEE 802.11e EDCA to provide more opportunity for real time traffic to access the channel depending upon their average queue occupancy. This will improve the uplink and downlink performance of the time constraint real time data traffic by reducing the packet drop rate, packet delay and improving the throughput.

According to our mechanism, the modified random early detection mechanism is implemented with high priority queues (Voice and Video) like AC[0] and AC[1] in all the stations including AP (Access Point). This will avoid the packet drop rate and packet delay of the uplink and downlink real time data traffic. According to the equation of Random Early Detection (RED) [1] as shown in (2):

$$Q_{avg} = (1 - W_q) \times Q_{avg} + Q_{inst} \times W_q \quad (2)$$

Where, W_q denotes Weight of the moving average function, Q_{avg} denotes Average Queue Occupancy and Q_{inst} denotes the Instantaneous Queue Occupancy.

The “threshold” of the average queue length is selected according to the priority of the traffic category. Since voice has higher priority, its threshold should be less than the threshold of the video traffic. That is,

$$threshold\ of\ voice\ queue < threshold\ of\ video\ queue \quad (3)$$

Figure 2 illustrates our proposed algorithm with respect to voice queue. This algorithm calculates the average voice

queue occupancy “avg” using the equation of RED. If the average queue occupancy is greater than the minimum and less than the maximum threshold of the voice queue, then the CW_{max} of the other traffic categories will be increased by the multiple factor of n_1 . And if the average queue occupancy is greater than the maximum threshold of the voice queue, then the CW_{max} of the other traffic categories will be increased by the multiple factor of n_2 . The n_1 and n_2 are the random values chosen in the defined interval, where n_2 is higher than n_1 . But, if the average queue occupancy is less than the minimum threshold of the voice queue, then the CW_{max} of the other traffic categories will be reset to their original values. The following flow chart illustrates our algorithm in detail.

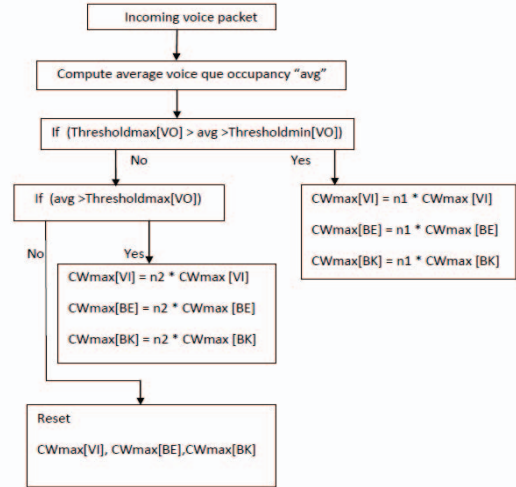


Figure 2. Modified random early detection implementation with AC[0] voice queue in IEEE 802.11e EDCA

Similarly, the contention adaptation algorithm is implemented in video queue and Fig.3 illustrates our algorithm in detail.

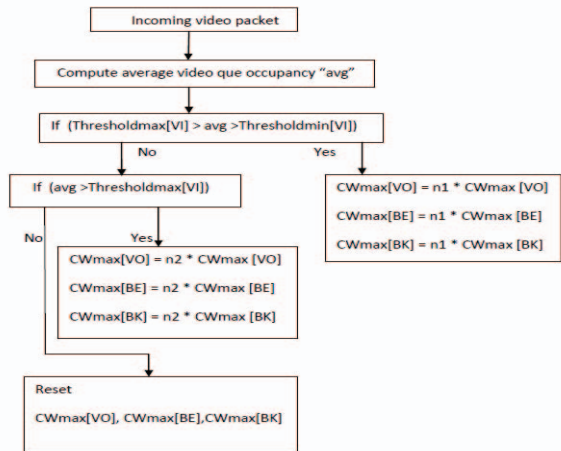


Figure 3. Modified random early detection implementation with AC[1] video queue in IEEE 802.11e EDCA

IV. SIMULATION PARAMETERS AND PERFORMANCE EVALUATION

A. Network Topology and System Simulation Parameters

We implement our proposed adaptive channel access mechanism for real time data traffic with IEEE 802.11e EDCA in network simulator 2 (NS-2). The simulation topology is depicted in Fig. 4.

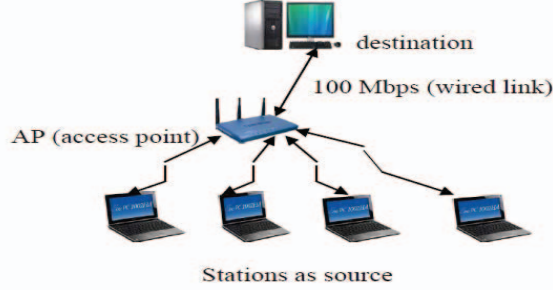


Figure 4. Simulation Network Topology

We adopt our methodology with IEEE 802.11e EDCA as the MAC layer protocol and use IEEE 802.11b, which supports 5.5 GHz and 11 Mbps as the wireless medium. We assume that all stations are in equal distance from the AP, the data rates of all the stations are identical. The parameters used in our simulations are illustrated in Table 1.0-1.2.

TABLE 1.0 SYSTEM PARAMETERS FOR SIMULATION

Parameters	Value
Simulator	NS2-2.28
MAC	IEEE 802.11e
Simulation Time	100 seconds
Number of Wireless Stations	1-15
Number of Access Point	1
Number of Wired Nodes	1
PHY	DSSS
Bandwidth	11 Mbps
Slot Time	20 μ s
SIFS	10 μ s
CW_{min} [AC_VO]	7 slots
CW_{max} [AC_VO]	15 slots
CW_{min} [AC_VI]	15 slots
CW_{max} [AC_VI]	31 slots
CW_{min} [AC_BE]	31 slots
CW_{max} [AC_BE]	1023 slots
CW_{min} [AC_BK]	31 slots
CW_{max} [AC_BK]	1023 slots
AIFS [AC_VO]	50 μ s
AIFS [AV_VI]	50 μ s
AIFS [AV_BE]	70 μ s
AIFS [AV_BK]	150 μ s

TABLE 1.1: TRAFFIC PARAMETERS FOR SCENARIO 1 & 2

Type	Distribution	Rate	Packet Size
Voice	Exponential	64 Kb/s	200 Bytes
Video	Constant Bit Rate	240 Kb/s	400 Bytes
Best Effort	Exponential	128Kb/s	200 Bytes
Back Ground	Exponential	128Kb/s	200 Bytes

TABLE 1.2: PARAMETERS FOR ADAPTIVE CHANNEL ACCESS MECHANISM IN SCENARIO 1 & 2

Parameters	Value
Queue length (packets)	50
Weighted Average (W_q)	0.1
Maximum Queue Threshold for voice (packets)	30
Minimum Queue Threshold for voice (packets)	20
Maximum Queue Threshold for video (packets)	45
Minimum Queue Threshold for video (packets)	40
n_1	Random (0, 1.5)
n_2	Random (1.5, 2)

B. Simulation Scenario & Performance Evaluation Metrics

We carry out our simulation in two different scenarios:

- **Scenario 1:** Two Audio + One Video traffic flows per wireless station.
- **Scenario 2:** Three Audio + One Video traffic flows per wireless station.

We perform simulations under different traffic load conditions and compare their performance of voice and video traffic in both uplink and downlink with standard IEEE 802.11e EDCA using the following performance evaluation metrics:

- End-to-End Packet Delay
- Packet Loss Rate
- Throughput

V. SIMULATION RESULTS AND DISCUSSION

A. Scenario I: 2 Audio+1 Video Traffic Flows/Station.

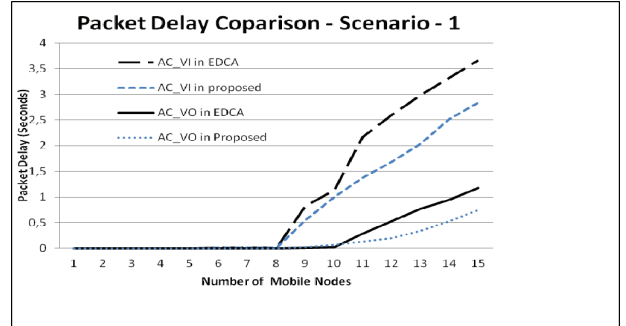


Figure 5. End-to-End Packet Delay Comparison

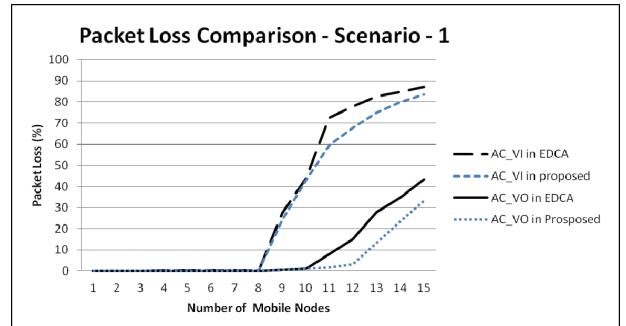


Figure 6. Packet Loss Rate Comparison

From Fig. 5 and 6, it is apparent that the end-to-end packet delay and packet loss rate for video (AC_VI) and voice (AC_VO) traffic of our proposed mechanism are lower than the existing channel access mechanism in EDCA. There is no packet delay or packet loss until the 8th mobile station, but they drastically increase after that. Our proposed algorithm reduces approximately the end-to-end packet delay by 0.75 second and packet loss rate by 10 % on average for video traffic and packet delay by 0.5 second and packet loss rate by 15 % on average for voice traffic under high traffic load.

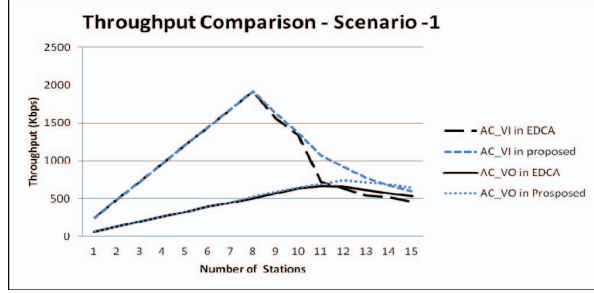


Figure 7. Throughput Comparison

From Fig. 7, it is obvious that the throughput starts decreasing after the 8th mobile station connected to the network due to the increase in packet loss. However, our proposed algorithm improves the throughput by 50% on average for video (AC_VI) and 15% on average for voice (AC_VO) comparing to the standard EDCA channel access mechanism under high traffic load.

B. Scenario II: 3 Audio+1 Video Traffic Flows/Station

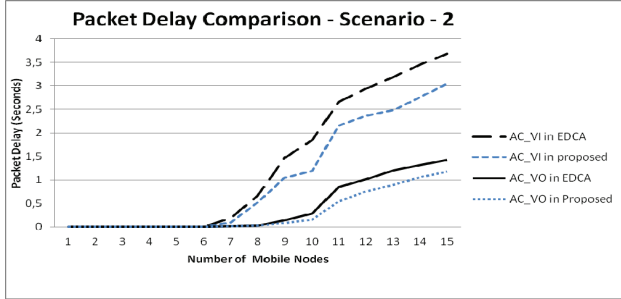


Figure 8. End-to-End Packet Delay Comparison

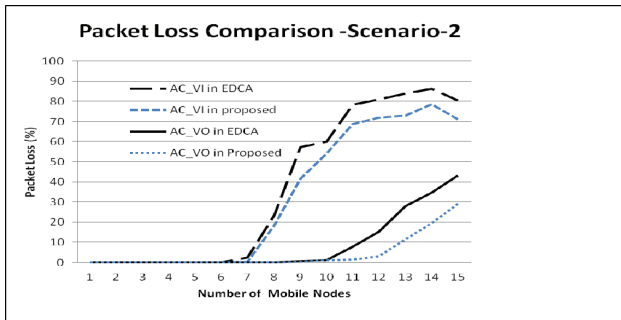


Figure 9. Packet Loss Rate Comparison

From Fig. 8 and 9, it is clear that the end-to-end packet delay and packet loss rate for video (AC_VI) and voice (AC_VO) traffic achieved from our proposed algorithm are lower than the existing channel access mechanism in EDCA. There is no packet delay or packet loss until the number of mobile stations is 6, but they drastically increase after that. Our proposed algorithm reduces approximately the end-to-end packet delay by 0.5 second and packet loss rate by 7 % on average for video traffic and packet delay by 0.3 second and packet loss rate by 10 % on average for voice traffic under high traffic load.

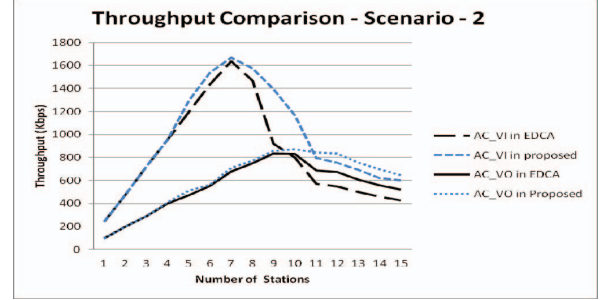


Figure 10. Throughput Comparison

From Fig. 10, we can observe that the throughput starts decreasing after the 7th mobile station connected to the network due to the increase in packet loss. However, our proposed algorithm improves the throughput by 15% on average for both video and voice traffic comparing to the standard EDCA channel access mechanism under high traffic load.

VI. CONCLUSIONS

In 802.11e EDCA, as the same service category in all the stations has the same EDCA parameters (AC[0] for voice which has the same CW_{min} , CW_{max} and TXOP for all the stations), there is a collision, which leads to performance degradation when all the stations have VC[0] traffic to transmit. In the scenario when all stations have video applications (VC[1] traffic category) and if some stations have more than one video flow, there will be delay in packet transmission in those stations having more flows because the EDCA parameters are identical for the particular traffic category irrespective of the number of flows, which leads to packet delay and performance degradation for real time applications.

In this research, we have tried to solve the problem of performance degradation of real time traffic under high traffic conditions. From all our simulation results, we can conclude that our proposed dynamic channel access mechanism improves the performance of voice and video traffic, especially under high traffic environment.

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